



Adaptive Human–Robot Interaction through Multimodal Emotion, Gesture, and Context Aware Policy Learning

An undergraduate project proposal report submitted to the

Department of Electrical and Information Engineering
Faculty of Engineering
University of Ruhuna
Sri Lanka

in partial fulfillment of the requirements for the

**Degree of the Bachelor of the Science of Engineering
Honours**

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Abstract

Human Robot Interaction is shifting from the command-based control systems to more complex systems. This is because the command-based systems are natural and intelligent communication. For robots to be able to successfully interact with human, they have to listen not only to spoken directions but also to human emotions, actions, and the surrounding environment. This project illustrates an adaptive Human Robot Interaction framework that allows robots to respond intelligently by learning from multimodal human cues.

The proposed system integrates emotion recognition, gesture and motion understanding, and context awareness using advanced techniques from Artificial Intelligence and Computer Vision. Unlike traditional rule-based systems, this model dynamically weights cues based on their reliability and the surrounding semantic context, allowing it to resolve scenarios where human cues are conflicting or ambiguous also.

By combining multimodal perception with context aware policy learning, the project contributes toward building "socially aware" robots that can interact with humans in a more empathetic, flexible and meaningful way. This approach aims to enhance the naturalness, safety and effectiveness of human robot collaboration in real world environments.

Contents

Acronyms	vi
1 Introduction	1
1.1 Background	1
1.2 Problem Statement	2
1.3 Objectives and Scope	2
2 Literature Review	4
2.1 Previous Work	4
2.1.1 HRI with rule-based systems	4
2.1.2 HRI with single modality	4
2.1.3 HRI with fusing more than one modality	5
2.1.4 HRI on edge devices	5
2.1.5 Available datasets for HRI research	6
2.2 Gaps in Literature	6
2.2.1 Lack of ambiguous social data	6
2.2.2 Fragmentation and Temporal Blindness in Intent Synthesis	6
2.2.3 Computational Bottleneck on Edge Robotics Platforms	6
2.2.4 Lack of Intent and Context-Aware Policy Learning	7
3 Methodology	8
3.1 Research Design	8
3.1.1 Multimodal Feature Extraction	9
3.1.2 Cross-Modal Fusion Architecture	9
3.1.3 Cross-Modal Fusion Architecture	11
3.1.4 Closed-Loop Sim-to-Real Tuning	13
3.1.5 Edge Optimization (TensorRT)	13
3.1.6 Multimodal "Golden Trial" Validation	13
3.2 Data Collection	13
4 Timeline and Resource Required	15
4.1 Timeline	15
4.2 Resource Required	15
5 Conclusion	17
References	17

List of Figures

3.1	System Architecture	8
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List of Tables

3.1	Comparison of Rule-Based vs. Fusion of Four Cues	11
4.1	Project Timeline	15
4.2	Required Hardware Resources	16
4.3	Required Software Resources	16

Acronyms

AI	Artificial Intelligence
AU	Action Unit
CUDA	Compute Unified Device Architecture
CV	Computer Vision
DDQN	Double Deep Q-Network
HRI	Human-Robot Interaction
LTS	Long Term Support
NLP	Natural Language Processing
RGB-D	Red Green Blue - Depth
ROS	Robot Operating System
VLM	Vision-Language Model
VRAM	Video Random Access Memory

Chapter 1

Introduction

Modern service robotics is currently grappling with a phenomenon known as **Behavioral Rigidity**, where machines fail to adapt their physical actions to the nuanced emotional and contextual signals inherent in human communication. Current systems are capable of detecting discrete human gestures, such as “pointing” at an exit, but it often fails to identify the meaning or the underlying intent of that gesture. For example, whether that “pointing” at an exit action is a helpful request for directions or an angry complaint regarding service. To address this problem, this project proposes a **Real-Time Multimodal Intent-Synthesis Framework**, followed by **intent and context-aware policy learning**. By fusing four different pillars of human communication, such as emotion, motion, gesture, and context, this system aims to resolve conflicting social cues and provide robotic responses that are not only physically safe but also socially aware and proactive.

1.1 Background

The evolution of human-robot interaction (HRI) has gone beyond simply performing a task. Robots should be able to interpret the “why” behind human actions. This helps robots to behave proactively rather than reactively. This evolution is driven by several key factors:

- **Dominance of Non-Verbal Communication:** Effective human communication heavily relies on a complex non-verbal signal, which includes facial micro-expressions, body language, and hand gestures. Academic research suggests that a significant portion of emotional intent is conveyed through these non-verbal channels, making social vision a prerequisite for empathetic robotics [1].
- **Challenge of Semantic Ambiguity:** A core problem in social robotics is semantic ambiguity, where a single gesture carries different meanings depending on the user’s emotional state and physical environment. For example, a person running towards a robot with a smile may symbolize a greeting, while the same movement with a frightened expression may indicate an emergency.

- **Technological Shift toward Transformers:** Traditional rule-based (IF–THEN) systems are considered too rigid to handle the ambiguous and conflicting nature of social data. Modern architectures are moving toward Transformer-based models. Transformer-based models utilize self-attention mechanisms to weigh different input streams simultaneously to resolve social dissonance.
- **Hardware and Edge-Compute Advancements:** The migration of complex “social brains” from remote servers to internal robotic platforms is enabled by advances in edge-computing hardware. Platforms like the NVIDIA Jetson Orin Nano, combined with optimization tools like TensorRT, allow for high-frequency processing. This is necessary for secure and responsive real-time interactions.

1.2 Problem Statement

Although there have been significant advances in individual sensing technologies, current HRI systems remain emotionally blind and behaviorally rigid. Most existing frameworks rely on fixed and rule-based policies, which fail to capture the dynamic nature of human behavior. This creates several critical challenges:

- **Ambiguity of Single Modalities:** A major limitation of current systems is their inability to resolve ambiguity when relying on a single source of information. For instance, a raised-hand gesture may represent a greeting, a command to stop, or an expression of surprise, depending on the user’s facial emotion and environmental context. Without multimodal fusion, robots frequently misinterpret human intent.
- **Inadequate Multimodal Fusion:** Existing systems often process emotions, gestures, and contextual cues as single inputs rather than as a fused signal. This leads to fragmented perception and a lack of holistic understanding.
- **Rigid Interaction Models:** Rule-based models often fail to adapt to human behavior in real time with varying human emotions or environmental contexts. This leads to unnatural or frustrating interactions between humans and robots.

Therefore, there is a clear need for a lightweight, multimedia HRI framework that jointly perceives emotional, gestural, and contextual cues to dynamically and efficiently adapt robot responses.

1.3 Objectives and Scope

Objectives

- **Synthesize a High-Fidelity Multimodal Dataset** – To generate a high-quality dataset comprising over 60 socially ambiguous interaction scenarios using NVIDIA Isaac Sim and Audio2Face, with automatic ground-truth annotations, supplemented by a limited real-world validation set.

- **Develop a Multimodal Fusion Engine** – To design and implement a late-fusion architecture that integrates emotional cues, physical gestures, human motion, and environmental context for robust intent inference.
- **Optimize for Edge Deployment** – To deploy and optimize the trained model on NVIDIA Jetson Orin Nano hardware using TensorRT, achieving real-time inference with latency below 100 milliseconds.
- **Validate in Simulation as well as with Real-World Scenarios** – Evaluate the proposed system in a simulation environment followed by some selected real-world scenarios.

Scope

This project focuses on the end-to-end design, integration, and deployment of a complete HRI system that enables socially intelligent robotic behavior. A key challenge addressed in this work is the lack of a unified dataset that jointly represents context, emotion, gesture, and motion, which are essential components of socially aware robot decision-making.

To address this limitation, the project emphasizes dataset integration and synthetic data generation, supported by self-supervised learning techniques, to bridge gaps between modality-specific datasets and enable coherent multimodal learning.

What is IN Scope:

- **Environments:** The study is limited to indoor service settings (specifically Hospitals, Museums, and Retail areas) where social norms are well-defined.
- **Human Cues:** The system will focus on non-verbal communication only. This includes facial micro-expressions (Affect), body language (Motion), and hand signals (Gesture).
- **Interaction Type:** One-on-one interactions between a single human and a single robot.

What is OUT of Scope:

- **Verbal Communication:** The project will not involve NLP or speech-to-text, focusing purely on visual/physical “Social Vision.”
- **Crowd Dynamics:** The model is not intended for “Social Navigation” in dense crowds (e.g., a busy train station); it focuses on direct interpersonal intent.
- **Outdoor Operation:** Weather-related variables, uneven terrain, and outdoor lighting fluctuations are excluded.
- **Mechanical Design:** The project uses off-the-shelf mobile bases (AgileX LIMO); it does not involve the design or fabrication of new robotic hardware.

Chapter 2

Literature Review

2.1 Previous Work

2.1.1 HRI with rule-based systems

Traditional HRI systems mostly rely on rigid, rule-based scripts to govern robot behavior. Therefore these systems typically use pre programmed "if-then" logic to map specific inputs to the outputs such as "if the user is happy, offer to play a game" [2]. However, this approach lacks the flexibility required for real world social interaction. Mamodiya et al. (2025) [3] note that these systems depend mainly on predefined scripts or single perceptual modalities. It makes them "brittle" when facial cues are occluded or lighting varies [4]. Furthermore, Ahmed and Aleem (2025) [5] shows that, as these robots are preprogrammed for specific tasks, they fail to recognize human emotions or communicate with contextual cues. It prevents them from adapting to changing environments.

2.1.2 HRI with single modality

Many existing HRI systems use single input type, which often leads to incomplete perceptions of the user's state.

- **Gestures Only:** Systems focus on gestures often struggle with environmental interference. As an example, the geometric relationship methods for the gesture recognition fail in noisy environments or complex backgrounds [6]. According to the subject differences in performing the same gesture like speed or range of motion, make it difficult for single-modality models to learn unseen feature variations to recognition failure [7]. But these systems have not ability to interpret intent or handle ambiguity without corroborating data.
- **Emotion Only:** Research relying only on emotion recognition faces significant robustness issues. Bamforth (2024) [8] notes that face based systems incorrect or fail when faces are blurred, not in the frame, or in low light. Lin et al. (2023) [9] further argue that main stream emotion recognition methods focus on short-term perception, disregarding the context in which emotions are perceived, which leads to identification failures. But these systems lack context and robustness against occlusion.

- **Motion Only:** Systems using only motion sensors or vision-based motion extraction can capture physical movement but miss the semantic meaning behind it. While effective for basic signaling, utilizing sensors can introduce increased communication costs and complexities, as individuals are required to wear sensor sets [10]. But motion data alone fails to capture the "why" behind an action (intent) or the emotional state of the user.

2.1.3 HRI with fusing more than one modality

To overcome single-modality limitations, recent research has focused on fusing multiple inputs.

- **ECRP** (Mamodiya et al., 2025) [3]: This framework fuses facial and vocal emotion cues using a confidence-weighted attention layer. It integrates these with a context vector (history, time-of-day) into a DDQN for adaptive task planning [11].
- **Transformer-Based Fusion** (Mohamed et al.) [12]: This approach utilizes modality-specific encoders for facial thermal data, facial AUs, and textual context. These are combined via additive fusion and processed by a shared transformer encoder to capture temporal dependencies.
- **FAM-HRI** (Lai et al., 2025) [13]: This system integrates gaze and speech data captured by wearable ARIA glasses. It uses a GPT-based agent to align spoken commands with gaze fixation periods to determine user intent in cluttered environments [14].
- **Context-Aware Two-Stream Network** (Wang et al., 2024) [10]: This method fuses first-person vision (video) with motion sensory data. One stream extracts visual context using attention maps (gaze), while the second processes motion data, combining them to recognize gestures for construction robots.
- **Hierarchical Fusion Engine** (Ahmed Aleem, 2025) [5]: This framework processes visual (facial), auditory (speech tone), and kinesthetic inputs. It uses a hierarchical fusion engine to integrate emotional and environmental data at feature, semantic, and decision levels to drive a collaborative decision engine.

2.1.4 HRI on edge devices

Deploying multimodal models on edge devices is critical for real-time interaction but presents computational challenges. Park et al. (2025) [15] utilized the NVIDIA Jetson AGX Orin to optimize open-vocabulary perception pipelines, using TensorRT to bridge the "real-time gap" and balancing accuracy with latency. Mamodiya et al. (2025) [3] optimized their ECRP framework for the NVIDIA Jetson Nano. They demonstrated that while Raspberry Pi 4 struggled (4 fps), the Jetson Nano with TensorRT acceleration achieved 30 fps and sub-400ms latency, proving the feasibility of complex fusion on low-power hardware [16].

2.1.5 Available datasets for HRI research

Data is essential for training adaptive HRI systems. The FER+ [3] and CK+ [8] datasets are widely used for facial emotion training, while RAVDESS [3] and IEMOCAP [9] are standard for vocal and multimodal emotion recognition. Chalearn is a large-scale RGB-D dataset for gesture recognition, containing both isolated and continuous gestures performed by multiple subjects. Widely used for vision-based hand/arm gesture recognition. For motion and action recognition, the NTU RGB-D [17] dataset is a common benchmark. To address the scarcity of real-world interaction data, simulations are frequently used. Mamodiya et al. (2025) [3] utilized a ROS-Gazebo environment to pre-train their reinforcement learning policy before fine-tuning it on real hardware.

2.2 Gaps in Literature

2.2.1 Lack of ambiguous social data

A significant gap that can be seen is the scarcity of unified datasets that capture all interaction pillars simultaneously. Most of the existing datasets focus more on pairs of modalities such as Audio-Visual rather than the full spectrum of Emotion, Gesture, Motion, and Context. Lin et al. (2023) [8] shows that the emotion recognition becomes more complex when "information conflicts arise between modalities," yet few datasets exist to train robots to resolve these conflicts effectively.

2.2.2 Fragmentation and Temporal Blindness in Intent Synthesis

Current research treats multimodal fusion as a secondary task which leads to temporal misalignment. Lai et al. (2025) [13] point out that only few methods model the "temporal alignment between gaze and speech," which note that the natural gaze behavior is noisy and users may look at an object moments before speaking. Viswanath et al. (2025) [18] argues that the robots often receive verbal commands (e.g., "put the cup there") that do not align perfectly in time with their visual perception. It suggest necessity of having a dedicated "Temporal Alignment module" to evaluate the relevance of inputs.

2.2.3 Computational Bottleneck on Edge Robotics Platforms

There is a difference between high accurate AI models and the mobile robot hardware capabilities. Park et al. (2025) [15] highlight this "real-time gap," note that while VLMs do the powerful recognition. Their high computational cost makes them difficult to deploy on edge devices like the Jetson Orin without optimization. This bottleneck forces a trade between semantic understanding and reaction speed [19].

2.2.4 Lack of Intent and Context-Aware Policy Learning

Most HRI systems focus on perception rather than action. Mamodiya et al. (2025) [3] argue robots can detect emotions, they often lack a "social policy" to adapt their behavior is a reason to "Behavioral Rigidity". Existing approaches often do not take "emotional context into consideration while planning for a task," result in robots that can sense a user is frustrated but cannot change dynamically their task planning to clear that frustration.

Chapter 3

Methodology

3.1 Research Design

This chapter presents the design science and experimental research approach, where an intelligent Human Robot Interaction system is designed, implemented, and evaluated. The research focuses on developing a multimodal adaptive interaction framework that enables a robot to understand:

- Human emotions
- Human gestures
- Human movements
- Interaction context

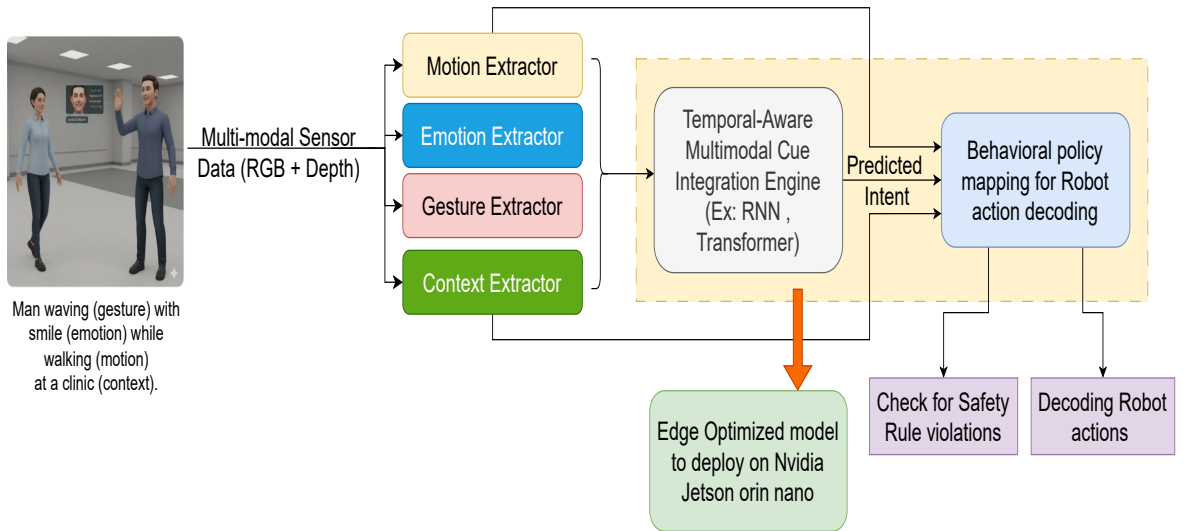


Figure 3.1: System Architecture

and to learn context aware interaction policies using AI and CV techniques.

3.1.1 Multimodal Feature Extraction

Four independent models are trained separately to capture key aspects of human behavior and environmental awareness. These models focus on human emotion recognition, human gesture recognition, human movement understanding, and interaction context identification. Training each model independently allows precise learning of each cue before higher-level fusion.

The raw image stream is processed by the robot using perception models that can work in real-time. Facial expression analysis is employed for extracting emotional expressions, while skeletal and hand pose models are used for gesture recognition. Human motion is analyzed by determining velocity and direction between frames, while environmental markers are detected for interaction context.

All the extracted cues are then converted to numerical feature vectors for angles, distances, intensity of movement, and spatial relations. Such a representation makes it easier to process visual data as it reduces complexities in visual data.

3.1.2 Cross-Modal Fusion Architecture

This is the heart of the process of understanding the intent, where all the four cues: human emotion, gesture, movement, and interaction context, are combined to understand what the user actually wants to do. All the cues are not considered to be of equal importance at all times. Rather, the system considers all the cues at once and gives more importance to those cues which best identify the current situation.

This fusion makes it possible for the system to cope with cases where human behavior is ambiguous or mixed. For example, a person can display a behavior of engagement through a gesture, but when he or she moves rapidly, it shows urgency. In this case, movement-related information will become more significant than gestures for determining the purpose.

Intent explains the reason behind a person's actions. While the perception stage identifies visible behaviors such as facial expressions, gestures, and movement, the intent layer focuses on understanding why those actions are happening and what is likely to happen next. This allows the robot to respond in advance instead of reacting only after a problem occurs.

Compared to rule based approaches, which rely on fixed conditions and respond only when a rule is triggered, intent based decision making allows the robot to behave more naturally and safely in real world human interactions. For example, When the emotion is angry and gesture is wave, due to their conflicting nature rule-based system gives an error while four-cues fused model identified it as frustrated or help seeking request from human. Also when the emotion is panicked

and the gesture is waving, the rule-based system ignores the panic emotion while through multi-modal fusion it is understood as emergency evaluation signal.

The Limitation of Rule-Based Systems in Social Robotics

In this subsection we explain some of the main limitations of rule-based systems and how our proposed fusion method will help to mitigate those limitations by using some practical scenarios.

1. The Complexity of Fuzzy Social Cues

Human social behavior is often unclear and strongly dependent on context. Emotions, gestures, movements, and situations rarely convey meaning on their own. Rule-based systems rely on fixed IF-THEN logic, which makes them poorly suited to handle such uncertainty.

The same action can represent different intentions depending on accompanying cues. For example, a smile with a hand wave may indicate a greeting, while a smile combined with fast movement may indicate urgency or stress. Handling these variations would require a large and complex set of rules, which becomes difficult to manage as the number of cues increases.

The Fusion of Four Cues approach addresses this limitation by jointly considering emotion, gesture, movement, and context. Instead of relying on predefined rules, the system evaluates all cues together to determine the most likely intent, allowing it to handle ambiguous social situations more effectively.

2. Handling Dissonant (Conflicting) Data

Humans frequently often give conflicting signals, and social behavior is not very easy to read. A rule-based strategy tends to stumble here, occasionally missing one cue or misreading how cues fit together. Take a grin combined with rapid movements-that can trip up a rigid rule system. The Fusion of Four Cues model repairs this by taking all emotion, gesture, movement, and context into consideration in concert. It does so by evaluating such factors in a manner that can identify the actual patterns-things like distinguishing an “Angry Wave” from a “Happy Wave.”

3. Temporal Dependencies

Human intention is not something static; it’s a process that takes time. Rule-governance systems fail to capture that, since their

scope is limited to the current state of observation. Picture a person waving their arm and approaching the robot with a visibly angry face. A system of rules could interpret both the wave and the movement towards the robot as a welcoming gesture despite an angry face. In turn, that could lead to a robot that reaches a person’s personal space.

The Fusion of Four Cues method addresses this issue in the combination of emotion, gesture, movement, and context in a temporal window. From the learning of this method, it was established that the anger signal is more weighted than the wave signal, and as a result, the robot will interpret the signal as “Threat/Avoidance.” With this knowledge, the robot will move slowly and maintain a safe distance.

Table 3.1: Comparison of Rule-Based vs. Fusion of Four Cues

Feature	Rule-Based	Fusion of Four Cues
Handling Noise	Rigid. If one sensor is slightly off, the whole rule breaks.	Robust. If the camera is blurry, it relies more on LiDAR or Motion.
Ambiguity	Fails. Cannot handle “Happy” and “Angry” at the same time.	Succeeds. It understands “Sarcasm” or “Dissonance” as a unique state.
Scalability	Impossible. A rule-based system would require 1,000,000 rules for every human variation.	Easy. It learns “Patterns” from 60+ Isaac Sim scenarios.
Timing	Static. It only looks at the “Current” frame.	Temporal. It remembers the path of the emotion over the last 5 seconds.

Table 3.1 further compares the suggested fusion method with rule-based in terms of noise and ambiguity handling capability, scalability and timing. So in summary, we hypothesize effective fusing of the desired four cues has the potential for ambiguity handling, temporal memory, global context awareness and hence can mitigate the rule-based system drawbacks.

3.1.3 Cross-Modal Fusion Architecture

Once the intent is understood, this layer converts that “Why” into a “How” using Proxemic Rules. It updates the robot’s navigation map by creating a large “in-

visible bubble” around the person that the robot should not enter, using ROS2 Behavior Trees. This step ensures that the robot not only avoids hitting the person but also maintains a distance that feels comfortable and respectful, turning AI decisions into socially acceptable actions.

Step A: Action Decoding

The fusion of four cues generates a probability distribution (e.g., 85% Emergency Passage). The intent decoder is the software component that selects the highest probability and determines: "Current Intent = EMERGENCY."

Step B: Behavioral Policy Mapping

This stage requires more than just intent recognition. The robot evaluates Intent + Context Decoder (environmental factors) to select an appropriate behavioral policy.

Case 1: Purpose is Emergency + Surroundings are Narrow Hallway
→ Policy: Yield position and move to wall.

Case 2: Intent is Emergency + Context is Open Lobby → Policy:
The social buffer distance should be increased and the present trend should be maintained.

Step C: Controller Execution

The chosen behavioral policy is sent to the ROS2 Nav2 Stack, where the calculations for the actual wheel velocities are done.

Example: The “Sarcastic Greeting”

Scenario: The Hostile Approach

Initial Classification: “Friendly Approach” identified (Individual displays wave gesture and move forward towards robot).

Decision Using Intent Only: The robot recognizes the “Approach”. It is the intent and causes the initiation of the forward motion to decrease the separation distance for in interaction initiation (for instance, in order to convey information or offer help).

Critical Failure: The robot ignored the conflicting emotional cues (hostile facial expression) and boundary of space principles. It did not to acknowledge that the person was displaying aggressive/confrontational behavior. By closing the distance, the robot violated the “Threat Response Zone” for this person. The upset person felt “threatened” because he/she could see the robot closing in on the robot’s approach as intrusive, responds with physical aggression to the robot.

The Need for Proxemic Integration: Intent classification identifies the behavioral goal; Proxemic analysis determines the socially appropriate interaction distance based on the detected emotional state.

3.1.4 Closed-Loop Sim-to-Real Tuning

In this section, the approach is using improvement rather than training once. The model is tested within the simulation environment. The issues that arise are carefully observed, including failing to recognize a gesture. The simulation parameters are varied, including light settings, speed, and noise on the sensor. This is done until the model is functioning properly within all scenarios.

This part of the work helps bridge the difference between simulations and reality. Simulations that are too simple for models do not work well on real-world robots. By gradually changing the simulated environment closer to real-world conditions, the learned behavior can be transferred to the real-world LIMO robot with little loss of performance.

3.1.5 Edge Optimization (TensorRT)

After this training process, the Transformer model's speed and size are optimized using NVIDIA TensorRT. This step eliminates unnecessary components, optimizes calculations for Jetson platforms, and adopts faster data representations with equal accuracy. It is very important for safety reasons because a slow processing speed can cause the robot to respond late. The model can now run in real time on a small computer on board thanks to this optimization.

3.1.6 Multimodal "Golden Trial" Validation

In this stage, the real world tests are conducted. People interact in the real world according to different behavior types, including cooperative behavior, goal-oriented movement, mixed behavior, and emergency behavior. This is where the robot's understanding of a human's intention is compared to the actual intention of the human. This stage demonstrates that the system is functional not only in a simulated environment but also in a real world environment.

3.2 Data Collection

- Multimodal Dataset Integration:
 - **Context Model:** HARWE dataset, which is used to understand various types of social and environmental contexts.

- **Emotion Model:** RAF-DB and IEMOCAP datasets for emotion Facial and Vocal Affect.
 - **Gesture Model:** ChaLearn and BEAT datasets for understanding human gestures.
 - **Motion Model:** NTU RGB-D dataset for tracking movement and predicting action.
- Create a dataset involving the generation of synthetic video clips that replicate different human behavior aspects, such as context, emotion, actions, and body movements, which would form the basis for training the four specific models. Isaac Sim would be used to create a realistic environment for controlling the different virtual scenarios.
 - Around 200-300 scenarios will be created in real-world indoor settings in order to test the models on real-world scenarios. The data will be captured using cameras in order to create a custom dataset that will be used for validating the models.

Chapter 4

Timeline and Resource Required

4.1 Timeline

The following table outlines the project timeline, detailing the phases from project selection to final documentation.

Table 4.1: Project Timeline

Tasks	Dec				Jan				Feb				Mar				Apr				May				Jun				Jul				Aug			
	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4				
Grouping and Project Selection																																				
Background Study																																				
Proposal Report Sub- mission																																				
Dataset Generation																																				
Multimodal Feature Extraction																																				
Cross-Modal Fusion Architecture																																				
Behavior Policy Map- ping																																				
Training and Verifica- tion (Simulated Data)																																				
Verify with Real World Data																																				
Edge Optimization (Jetson Nano)																																				
Testing and Finaliza- tion																																				
Final Documentation and Paper																																				

4.2 Resource Required

This section outlines the hardware and software resources required for implementing the multi-cue human intent recognition system with synthetic data generation and real-world validation.

Table 4.2: Required Hardware Resources

Component	Specifications	Purpose	Qty
NVIDIA RTX 3090	24GB VRAM, CUDA-enabled	Synthetic data generation (Isaac Sim), model training	1
NVIDIA Jetson Orin Nano	8GB RAM, edge AI computing	Real-world deployment and validation	1
Intel RealSense D455	RGB-D camera with depth sensing	Real-world data capture (emotion, gesture, motion, context)	1

Table 4.3: Required Software Resources

Software	Version/Details	Purpose
NVIDIA Isaac Sim	Latest version with Omniverse	Synthetic data generation for multi-cue scenarios (distance, proximity, speed, human interactions)
Ubuntu	22.04 LTS	Operating system for development and deployment
ROS2	Humble Hawksbill (recommended for Ubuntu 22.04)	Robot middleware for sensor integration, data processing, and system communication

Chapter 5

Conclusion

This project proposes a **Real-Time Multimodal Intent-Synthesis Framework** to address the” **Behavioral Rigidity**” problem of the modern service robots. By integrating four primary non-verbal cues of humans such as Emotion, Motion, Gesture, and Context, this system moves beyond simple obstacle avoidance toward a deep understanding of human social intent. The technical core of this proposal lies in the shift from rigid rule-based logic to late-fusion transformer architecture. This shift allows the robot to use Attention Mechanisms to solve the problems of” **Social Dissonance**” and” **Semantic Ambiguity**”. Also, it allows to weight conflicting signals such as an angry face paired with a” requesting” gesture to derive a socially appropriate response. Furthermore, this project mitigates the” **Reality Gap**” by utilizing NVIDIA Isaac Sim and Audio2Face to synthesize high-fidelity datasets with automated ground-truth labeling. This ensures that the model is trained on complex scenarios that are difficult to capture in the real world. The importance of this research is emphasized by its focus on Edge-Native Deployment. By optimizing the framework with TensorRT, the project ensures that a high-level social intelligence system can run natively on an NVIDIA Jetson Orin Nano with latency below 100ms. This makes the robot responsive and safe in high-end environments such as hospitals and retail stores. Ultimately, this framework provides the” **Social Brain**” for service robots to transition from reactive machines to proactive, intent-aware collaborators. To ensure the performance of these models is grounded in reality, the project culminates in a Real-World” **Golden Trial**” validation. By testing the framework against physical data from real human volunteers, we will get the definitive metrics needed to prove the system is reliable in the unpredictable ”**Chaotic Reality**” of human interactions. This allows to effectively bridge the final gap between simulation and real-world deployment.

References

- [1] M. Argyle, *Bodily Communication*. London: Methuen, 2 ed., 1988.
- [2] F. Alonso-Martin, M. Malfaz, J. Sequeira, J. F. Gorostiza, and M. A. Salichs, “A multimodal emotion detection system during human–robot interaction,” *Sensors*, vol. 13, no. 11, pp. 15549–15581, 2013.
- [3] U. Mamodiya, I. Kishor, A. A. Syed, P. Sankalkar, and N. Naik, “An adaptive human-robot interaction framework using real-time emotion recognition and context-aware task planning,” *IEEE Access*, 2025.
- [4] Z. Su and W. Sheng, “Context-aware conversation adaptation for human-robot interaction,” in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 2542–2548, IEEE, 2024.
- [5] A. Ahmed and S. Aleem, “Enhancing human-robot collaboration through multimodal emotion and context-aware,” *Available at SSRN 5237859*, 2025.
- [6] M. Hu, “Research on human-robot interaction technology based on gesture recognition,” *Journal of Electronic Research and Application*, vol. 9, no. 6, 2025.
- [7] H. Chen, M. C. Leu, and Z. Yin, “Real-time multi-modal human–robot collaboration using gestures and speech,” *Journal of Manufacturing Science and Engineering*, vol. 144, no. 10, p. 101007, 2022.
- [8] J. Bamforth, *Multimodal Emotion Recognition Software to Facilitate Human-Robot Interaction*. PhD thesis, Sheffield Hallam University, 2025.
- [9] Z. Lin, F. Cruz, and E. B. Sandoval, “Self context-aware emotion perception on human-robot interaction,” *arXiv preprint arXiv:2401.10946*, 2024.
- [10] X. Wang, D. Veeramani, F. Dai, and Z. Zhu, “Context-aware hand gesture interaction for human–robot collaboration in construction,” *Computer-Aided Civil and Infrastructure Engineering*, vol. 39, no. 22, pp. 3489–3504, 2024.
- [11] Y. Yano, A. Mizutani, Y. Fukuda, D. Kanaoka, T. Ono, and H. Tamukoh, “Unified understanding of environment, task, and human for human-robot interaction in real-world environments,” in *2024 33rd IEEE International Conference on Robot and Human Interactive Communication (ROMAN)*, pp. 224–230, IEEE, 2024.

- [12] Y. Mohamed, S. Lemaignan, A. Guneyusu, P. Jensfelt, and C. Smith, “Fusion in context: A multimodal approach to affective state recognition,” *arXiv preprint arXiv:2409.11906*, 2024.
- [13] Y. Lai, S. Yuan, B. Zhang, B. Kiefer, P. Li, T. Deng, and A. Zell, “Fam-hri: Foundation-model assisted multi-modal human-robot interaction combining gaze and speech,” *arXiv preprint arXiv:2503.16492*, 2025.
- [14] J. Engel, K. Somasundaram, M. Goesele, A. Sun, A. Gamino, A. Turner, A. Talattof, A. Yuan, B. Souti, B. Meredith, *et al.*, “Project aria: A new tool for egocentric multi-modal ai research,” *arXiv preprint arXiv:2308.13561*, 2023.
- [15] J. Park, P. Kim, and D. Ko, “Real-time open-vocabulary perception for mobile robots on edge devices: a systematic analysis of the accuracy-latency trade-off,” *Frontiers in Robotics and AI*, vol. 12, p. 1693988, 2025.
- [16] T. Kim, E. Moon, H. Kang, and H. S. Kim, “Omer-npu: on-device multimodal emotion recognition on neural processing unit for low latency and power consumption,” *Neural Computing and Applications*, pp. 1–28, 2025.
- [17] W. Zhao, K. Gangaraju, and F. Yuan, “Multimodal perception-driven decision-making for human-robot interaction: a survey,” *Frontiers in Robotics and AI*, vol. 12, p. 1604472, 2025.
- [18] A. Viswanath, L. Veeramacheneni, and H. Buschmeier, “Enhancing explainability with multimodal context representations for smarter robots,” *arXiv preprint arXiv:2503.16467*, 2025.
- [19] C. V. Reddy, N. U. Reddy, S. Sharma, J. P. Kumar, S. Lakhanpal, and A. Albawi, “Advanced deep learning architectures for real-time human emotion recognition and behavioral prediction,” in *2024 International Conference on Communication, Computer Sciences and Engineering (IC3SE)*, pp. 1306–1311, IEEE, 2024.